MIL-R-2341F
27 April 1984

SUPERSEDING
MIL-R-2341E
27 July 1978

MILITARY SPECIFICATION

ROD AND CLAMP, INSECT BAR

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document covers one type and size of rod and clamp for holding insect bars (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents. Unless otherwise specified, the following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this document to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-W-428	-	Wire, Steel, Carbon (High Carbon, Round For Mechanical Springs, General Purpose)
TT-C-490	-	Cleaning Methods and Pretreatment of Ferrous Surfaces for Organic Coatings
TT-E-529	-	Enamel, Alkyd, Semi-Gloss
UU-P-268	-	Paper, Kraft, Wrapping
PPP-B-566	-	Boxes, Folding, Paperboard
PPP-B-636	-	Boxes, Shipping, Fiberboard
PPP-B-676	-	Boxes, Setup
PPP-T-45	-	Tape, Gummed, Paper Reinforced and Plain, for Sealing and Securing

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Natick Research and Development Center, Natick, MA 01760 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 7210

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STANDARDS

FEDERAL

FED-STD-595 - Colors

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-130 - Identification Marking of U.S. Military Property
MIL-STD-147 - Palletized Unit Loads

DRAWINGS

U.S. ARMY NATICK RESEARCH AND DEVELOPMENT CENTER

2-5-45 - Rod and Clamp, Insect Bar; Assembly and Details
2-5-49 - Rod and Clamp, Insect Bar; Rod Assembly and Details
2-5-61 - Rod and Clamp, Insect Bar; Mounting Methods

(Copies of documents required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. Unless otherwise specified, the following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this document to the extent specified herein.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A 108 - Steel Bars, Carbon, Cold Finished, Standard Quality
A 366 - Steel, Carbon, Cold-Rolled Sheet, Commercial Quality
A 575 - Merchant Quality Hot-Rolled Carbon Steel Bars
B 633 - Electrodeposited Coatings of Zinc on Iron and Steel
D 3951 - Standard Practice for Commercial Packaging
E 527 - Numbering Metals and Alloys (UNS)

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Technical society and technical association documents are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence.

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3. REQUIREMENTS

3.1 First article. When specified, a sample shall be subjected to first article inspection (see 4.3, 6.2, and 6.4).

3.2 Materials (see 6.3). Materials shall conform to the applicable documents and requirements specified herein.

3.2.1 Steel, sheet and strip. The steel sheet and strip shall conform to chemical composition of ASTM A 366 (UNS G10080-G10020 of ASTM E 527). The sheet material shall be hot or cold rolled commercial quality. The steel strip shall be hot-rolled commercial quality of cold-rolled tempers of ASTM A 366.

3.2.2 Steel bars, carbon. The carbon steel bars shall conform to grades 101 through 1020 of ASTM A 108 (UNS G10100-G10200 of ASTM E 527); or 1008 through 1020 or M1008 through M1020 of ASTM A 575 (UNS G10080-G10200 of ASTM E 527).

3.2.3 Primer. The primer shall conform to TT-P-636.

3.2.4 Enamel. The enamel shall conform to class B of TT-E-529. The color shall be Olive Drab No. 24087 of FED-STD-595.

3.2.5 Coatings. As an alternate to the primer and enamel specified in 3.2.3 and 3.2.4, the rod and insect bar may be coated with standard Corvell ECA-1555-FC3 Epoxy Coating, Duracron Super 600, or equal. The coating shall be applied in accordance with the manufacturer's instructions. Prior to use of the "or equal" item, the contractor shall submit the item to the contracting officer with supporting data for approval.

3.3 Construction. The construction of the rod and clamp shall conform to the requirements specified herein and to the dimensions shown on Drawings 2-5-45 and 2-5-49.

3.3.1 Clamp assembly. The clamp assembly shall conform to requirements as shown on Drawing 2-5-45.

3.3.1.1 Body and clip. The body and clip shall be fabricated from material specified in 3.2.1.

3.3.1.2 Bolt. The bolt shall be fabricated from material specified in 3.2.2.

3.3.1.3 Wing nut. The wing nut shall be fabricated from material specified on Drawing 2-5-45.

3.3.2 Rod assembly. The rod assembly shall conform to Drawing 2-5-49.

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3.3.2.1 Upright and crossbar. The upright and crossbar shall be fabricated form material specified in 3.2.2. The crossbar shall be centered on the upright.

3.3.2.2 Ring. The ring shall be fabricated from type II, class 1 or 2, 0.020 inch diameter steel wire conforming to QQ-W-428. The ring shall be constructed as shown on Drawing 2-5-49.

3.4 Finish.

3.4.1 Clamps. After fabrication, the clamp body, clip, bolt, and wing nut shall be thoroughly cleaned and electro-zinc plated and chromated in accordance with type II, SC 3 of ASTM B 633 except that the salt spray requirement does not apply.

3.4.2 Rods.

3.4.2.1 Preparation of surfaces. All rod surfaces shall be prepared for painting by any method, type I or II treatment of TT-C-490 except preproduction approval of type I is waived.

3.4.2.2 Application of primer. The prepared rod surfaces shall be given one coat of metal primer specified in 3.2.3, and thoroughly dried prior to application of the enamel.

3.4.2.3 Application of enamel. The primed surfaces of the insect bar rod shall be coated with enamel specified in 3.2.4 and baked. The enamel finish shall level out to produce a smooth and uniform surface without objectionable runs, wrinkles, areas of no film, orange peel, grit, and flooding. Crossbar rod assemblies and the upright rods shall be coated separately, precaution shall be taken to keep the coating from interfering with the assembly or disassembly of the rods at the T-joint.

3.5 Instruction sheet. An instruction sheet illustrating the methods of mounting the insect bar clamps on various types of beds, as shown on Drawing 2-5-61, shall be packed with each box of four clamps.

3.6 Marking for identification. Identification marking shall be in accordance with MIL-STD-130. The letters "U.S." and the manufacturer's identification shall be indented prior to finishing on each clamp assembly, crossbar or crossbar coupling, and upright of each insect bar rod assembly.

3.7 Workmanship. The finished rod and clamp shall conform to the quality of product established by this document. All sharp edges and corners shall be broken.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the document where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

4.3 First article inspection. When a first article is required (see 6.2), it shall be examined for the defects specified in 4.4.3 and 4.4.4. The presence of any defect shall be cause for rejection of the first article.

4.4 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this document or applicable purchase document.

4.4.2 In-process inspection. In-process inspection shall be conducted to see that accomplishment of the following is in accordance with the specified requirements. Whenever nonconformance is noted, correction shall be made to affected items in process.

a. Surface preparation. Cleaning and conditioning of metal surfaces for painting is accomplished as specified in 3.4.2.1.
b. Priming of metal. Metal surfaces have been coated with primer as specified in 3.4.2.2.

4.4.3 End item visual examination. The end item shall be examined for the defects listed in table I. The lot size shall be expressed in units of rod and clamp assemblies. The sample unit shall be one rod and clamp assembly. The inspection level shall be II and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 2.5 for major defects and 6.5 for total (major and minor combined) defects.

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TABLE I. End item visual defects

Examine	Defect	Classification Major	Minor
Finish of enameled, or epoxy coating (rod assembly)	Not finished	X	
	Wrong color		X
	Area of no film		X
	Not smooth, not uniform, or not adherent (i.e., contains wrinkles, runs, orange peel, grit, or color separation)		X
Finish of plated surfaces (clamp assembly)	Not plated	X	
	Plating not uniform		X
	Area of no plating	X	
	Dirt or foreign matter imbedded		X
Construction and workmanship	Any functioning component that is inoperative or requires undue force to operate	X	
	Ring missing or not as specified		X
	Any component missing (including bolt or wing nut) except ring	X	
	Component is not in proper alignment, broken, fractured, or split	X	
	Any sharp edge, burr, or sliver	X	
	Corners on clamp not rounded as indicated on drawing		X
	Hole is too large or too small	X	
	Any end of bar not chamfered as required		X
	Sharp corners not broken or rounded where required		X
Marking	Missing, incomplete, not legible, not specified type or size, not affixed in the prescribed manner or otherwise incorrect		X

End item dimensional examination. The end item shall be examined for conformance to the specified dimensions on the applicable drawings. Any dimension not within the specified tolerance shall be classified as a defect. The lot size shall be expressed in units of rod and clamp assemblies. The sample unit shall be one rod and clamp assembly. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 4.0.

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4.4.5 Packaging inspection. An examination shall be made to determine that the preservation, packing, and marking comply with the section 5 requirements. Defects shall be scored as specified in table II. The sample unit shall be one shipping container fully packaged with the exception that it need not be closed. Examination of closure defects listed in table II shall be made on shipping containers fully packaged. The lot size shall be the number of containers in the inspection lot. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 2.5.

TABLE II. Packaging defects

Examine	Defect
Marking (exterior and interior container)	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application.
Material	Any component missing (e.g., instruction sheet not packed with each box of four clamps), damaged, or not as specified.
Workmanship	Inadequate application of components such as: incomplete closure of container flap, loose strapping, improper taping, or inadequate stapling. Bulged or distorted container.
Contents (interior and exterior container)	Number is more or less than required.

4.4.6 Palletization inspection. An examination shall be made to determine that the palletization complies with the section 5 requirements. Defects shall be scored in accordance with the list below. The sample unit shall be one palletized unit load fully packaged. The lot size shall be the number of palletized unit loads in the end item inspection lot. The inspection level shall be S-1 and the AQL, expressed in terms of defects per hundred units, shall be 6.5.

Examine	Defect
Finished dimensions	Length, width, or height exceeds specified maximum requirements.
Palletization	Not as specified. Pallet pattern not as specified. Interlocking of loads not as specified. Load not bonded with required straps as specified.

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Examine	Defect
Weight	Exceeds maximum load limit.
Marking	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or Commercial as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Rod assembly. Each rod assembly shall be disassembled and the components nested and wrapped, sleeved, or bagged in 30-pound basis weight kraft paper conforming to type I, grade B of UU-P-268. The sleevel shall be of sufficient length to allow a minimum 2-inch fold over at each end. The wrap, sleeve, or bag shall be secured with 2-inch minimum width tape conforming to type III, grade B of PPP-T-45. Two uprights and two crossbars, wrapped or bagged as specified shall be nested in the most compact manner and secured with tape specified herein. Tape shall be applied approximately 10 inches from each end. Ten uprights and ten crossbars shall then be unit packed in a snug-fitting fiberboard box conforming to style FPF, type CF, variety optional, or type SF, class domestic, grade 200 of PPP-B-636. The contents shall be immobilized with cushioning material, or die cut or scored fiberboard sheets made form the same material as the box. Closure shall be in accordance with the appendix of the box specification.

5.1.1.2 Clamp assemblies.

5.1.1.2.1 Unit pack. Four clamp assemblies shall be disassembled and nested together in a snug-fitting folding or setup paperboard box. The folding paperboard box shall conform to variety 1, type III or VIII, type G, class 1 of PPP-B-566. The setup paperboard box shall conform to type I, variety 1, class C, style 4 of PPP-B-676. One copy of the instruction sheet (Drawing 2-5-61) shall be folded and placed in the box with the clamps. Closure of each setup or folding paperboard box shall be effected with tape specified in 5.1.1.1. The tape shall be centered girthwise around the box and overlap itself not less than 2 inches.

5.1.1.2.2 Intermediate pack. Forty clamp assemblies, unit packed as specified in 5.1.1.2.1, shall be packed in a snug-fitting intermediate fiberboard box conforming to style RSC, type CF, variety optional, or type SF, class domestic, grade 200 of PPP-B-636. Closure shall be in accordance with the appendix of the box specification.

5.1.2 Commercial. Rod and clamp assembly shall be preserved in accordance with ASTM D 3951.

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5.2 Packing. Packing shall be level A, B, or Commercial as specified (see 6.2).

5.2.1 Level A packing.

5.2.1.1 Rod assemblies. Twenty rod assemblies, preserved as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style FTC, V2s of PPP-B-636. Closure and reinforcing with steel strapping or tape banding shall be in accordance with the appendix of PPP-B-636, except that the inspection shall be in accordance with 4.4.5.

5.2.1.2 Clamp assemblies. One hundred and sixty clamp assemblies, preserved as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC, V2s of PPP-B-636. Each shipping container shall be closed in accordance with method III, waterproofed in accordance with method V, and reinforced as in the appendix of PPP-B-636, except that the inspection shall be in accordance with 4.4.5.

5.2.2 Level B packing.

5.2.2.1 Rod assemblies. Twenty rod assemblies, preserved as specified in 5.1, shall be packed in a snug-fitting double walled, fiberboard shipping container conforming to style FTC, type CF, class domestic, grade 275 of PPP-B-636. Each shipping container shall be closed and reinforced as specified in 5.2.1.1.

5.2.2.2 Clamp assemblies. One hundred and sixty clamp assemblies, preserved as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC, type CF, variety optional, or type SF, class domestic, grade 275 of PPP-B-636. Closure shall be in accordance with method II as specified in the appendix of PPP-B-636, except that the inspection shall be in accordance with 4.4.5.

5.2.2.3 Weather-resistant containers. When specified (see 6.2), the shipping container shall be grade V3c, V3s, or V4s fiberboard box fabricated in accordance with PPP-B-636 and closed in accordance with method III as specified in the appendix of PPP-B-636, except that the inspection shall be in accordance with 4.4.5.

5.2.3 Commercial packing. Rod and clamp assembly preserved as specified in 5.1, shall be packed in accordance with ASTM D 3951.

5.3 Palletization. When specified (see 6.2), rod and clamp, insect bars, packed as specified in (5.2), shall be palletized in accordance with load type I of MIL-STD-147. Each prepared load shall be bonded with primary and secondary straps in accordance with bonding means K and L. Pallet patterns shall be in accordance with the appendix of MIL-STD-147. Interlocking of

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loads shall be effected by reversing the pattern of each course. If the container is of a size which does not conform to any of the pallet patterns specified in MIL-STD-147, the pallet pattern used shall first be approved by the contracting officer.

5.4 Marking. In addition to any special marking required by the contract or purchase order, interior packs and shipping containers and palletized loads shall be marked in accordance with MIL-STD-129 or ASTM D 3951, as applicable.

6. NOTES

6.1 Intended use. The rod and clamp is intended for use in supporting insect bars over certain types of military beds and cots.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this document.
b. When first article sample is required (see 3.1, 4.3, and 6.4).
c. Selection of applicable levels of preservation and packing (see 5.1 and 5.2).
d. When weather-resistant grade fiberboard shipping containers are required for level B packing (see 5.2.2.3).
e. When palletization is required (see 5.3).

6.3 Recycled material. It is encouraged that recycled material be used when practical, as long as it meets the requirements of this document.

6.4 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of DAR 7-104.55. The first article should be a preproduction sample. The first article should consist of one rod assembly and one clamp assembly. The contracting officer should include specific instructions in all acquisition documents regarding arrangements for inspection, and approval of the first article.

6.5 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-R-2341F	Rod and Clamp, Insect Bar

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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NATICK OP, 1 Apr 84

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